1. System overview

Component	Detail
Application server	J2EE 1.3.1 Reference Implementation on Java 1.4, port 8000. Four applications: `/petstore` (Storefront), `/opc` (Order Processing Center), `/supplier`, `/admin`.
Messaging	JMS queues and topics inside the J2EE server: `jms/opc/OrderQueue`, `jms/opc/OrderApprovalQueue`, `jms/supplier/PurchaseOrderQueue`, `jms/opc/InvoiceTopic`, mail queues
Database	Cloudscape (Apache Derby), database `PetStoreDB` in `/var/petstore/db`, schema ESTOREUSER. Batch jobs connect through the network server on port 1527.
Batch home	`/opt/petstore`: `batch/` scripts, `db/batch/` SQL, `conf/` settings
Output folders	`/var/petstore/backup`, `/var/petstore/extract`, `/var/petstore/log`, `/var/petstore/tmp`
Job log	`/var/petstore/log/batch.log`, one line per job start and end

Java™ Pet Store 1.3.2

1.1 Order flow

Operations Runbook

```
Storefront --OrderQueue--> OPC PurchaseOrderMDB
   total under limit (US 500, JP 50000)  --> APPROVED automatically
   otherwise                              --> PENDING until the Admin client approves or denies
APPROVED --> Supplier (PurchaseOrderQueue) --> ships from INVENTORY --> InvoiceTopic --> OPC
OPC marks the order COMPLETED when every line is shipped
```

Document	PS-RB-01
Version	1.3.2
Audience	Operations and Control-M schedulers
Scheduler	Control-M, data center DC_PS_PRD, node ps-app01

2. Batch schedule

The schedule is defined in `scheduler/control-m/PS_PETSTORE.xml` (two folders).

Job	Folder	When	Waits for	Script
PS_DB_BACKUP	PS_PETSTORE_DAILY	Daily 01:00	-	ps_db_backup.sh
PS_ORDER_EXTRACT	PS_PETSTORE_DAILY	Daily 01:30	PS_DB_BACKUP-OK	ps_order_extract.sh
PS_ORDER_AGING_REPORT	PS_PETSTORE_DAILY	Daily 01:45	PS_ORDER_EXTRACT-OK	ps_order_aging_report.sh
PS_INVENTORY_REPORT	PS_PETSTORE_DAILY	Daily 01:50	PS_ORDER_EXTRACT-OK	ps_inventory_report.sh
PS_ORDER_PURGE	PS_PETSTORE_DAILY	Sunday 02:00	PS_ORDER_EXTRACT-OK	ps_order_purge.sh
PS_CATALOG_CHECK	PS_PETSTORE_DAILY	Daily 06:00	-	ps_catalog_check.sh
PS_HEALTH_CHECK	PS_PETSTORE_CYCLIC	Every 10 minutes, 00:00-23:59	-	ps_health_check.sh

```
01:00 PS_DB_BACKUP --> 01:30 PS_ORDER_EXTRACT --+--> 01:45 PS_ORDER_AGING_REPORT
                                                +--> 01:50 PS_INVENTORY_REPORT
                                                +--> 02:00 PS_ORDER_PURGE (Sundays)
06:00 PS_CATALOG_CHECK            every 10 min  PS_HEALTH_CHECK
```

3. Job details

3.1 PS_DB_BACKUP

Online backup of PetStoreDB with `SYSCS_UTIL.SYSCS_BACKUP_DATABASE` (`db/batch/db_backup.sql`). The backup is moved to `/var/petstore/backup/PetStoreDB_YYYYMMDD`. Backups older than 7 days are deleted. Critical job: up to 2 reruns 10 minutes apart. On failure Control-M mails ps-ops and PS_ORDER_EXTRACT stays waiting.

3.2 PS_ORDER_EXTRACT

Exports yesterday's purchase orders (00:00 to 24:00 server time) to `/var/petstore/extract/orders_YYYYMMDD.csv` for Finance (`db/batch/order_extract.sql`).

Column	Content
1 po_id	Order number, from the 1001 counter
2 po_user_id	User name of the customer
3 po_date	Order time, epoch milliseconds
4 po_locale	en_US, ja_JP or zh_CN. The order value is in the currency of the locale.
5 po_value	Order total
6 lines	Number of order lines
7 status	PENDING, APPROVED, DENIED or COMPLETED

No header row. The last line is a trailer `TRAILER,YYYYMMDD,<row count>`. Failure mails ps-finance.

3.3 PS_ORDER_AGING_REPORT

Lists orders that are still PENDING or APPROVED after 24 hours, and orders the supplier has invoiced but the OPC has not completed, in `/var/petstore/extract/order_aging_YYYYMMDD.txt`. Exit code 4 when any is listed; Control-M mails ps-ops.

3.4 PS_INVENTORY_REPORT

Lists supplier items with stock below 500 units in `/var/petstore/extract/inventory_YYYYMMDD.txt`.

3.5 PS_ORDER_PURGE

Sundays only. Deletes COMPLETED and DENIED orders older than 180 days, with their order lines, supplier orders, contact information, addresses and payment cards. It waits for that day's extract so no order is deleted before it is extracted. PENDING and APPROVED orders are never purged.

3.6 PS_CATALOG_CHECK

Runs catalog checks C1 to C5 (`db/batch/catalog_check.sql`): category without product, product without item, item priced below cost, item missing a description in a locale the store serves, item the supplier does not stock. If the catalog tables do not exist it calls `http://ps-app01:8000/petstore/Populate?forcefully=false` to load the sample data, then checks again. Exit code 4 means a check found rows; Control-M mails ps-catalog. Details are in `/var/petstore/log/catalog_check.log`.

3.7 PS_HEALTH_CHECK

Every 10 minutes. Requests the Storefront main screen, the OPC `EntityCatalog.jsp`, and the Supplier and Admin entry pages. Exit code 8 when any does not answer HTTP 200; Control-M mails ps-ops.

4. Exit codes and recovery

Code	Meaning	Action
0	OK	None
4	A report or check found something: stuck orders (PS_ORDER_AGING_REPORT) or bad catalog data (PS_CATALOG_CHECK)	Read the report. For stuck PENDING orders ask the order administrator to process them. For catalog data, fix the data and rerun.
8	SQL, file or HTTP error	Read `batch.log` and the job log. Check that the J2EE server is up and that port 1527 answers. Rerun.

4.1 Restore from backup

1. Stop the J2EE server and Cloudscape.
2. Move `/var/petstore/db/PetStoreDB` aside.
3. Copy `/var/petstore/backup/PetStoreDB_YYYYMMDD` to `/var/petstore/db/PetStoreDB`.
4. Start Cloudscape and the J2EE server, then open the Storefront home page.

4.2 Rerunning the extract

The extract always covers the day before the run date. To re-extract an older day, run the export query in `db/batch/order_extract.sql` by hand with the two time limits changed.

4.3 Stuck APPROVED order

An order stays APPROVED when the supplier's invoice message never reaches the OPC (listed as MISMATCH in the aging report). Resubmitting the supplier inventory in the Supplier client makes the OPC re-evaluate unfulfilled orders.

5. Settings

```
conf/env.sh              PS_HOME, folders, PS_DB_URL, retention, thresholds, ps_ij and ps_log
helpers
conf/petstore.properties database, storefront, OPC, JMS, supplier and batch settings
```

Setting	Value
Order retention	180 days
Backup retention	7 days
Stuck order limit	24 hours
Low stock limit	500 units
Database URL for batch	`jdbc:derby://localhost:1527//var/petstore/db/PetStoreDB`